Curriculum Intelligence Engine — Week 8: Final Report

Group 9 · Roll Nos: J048, J049, J050, J052, J053 · AI University NLP Course Project 2026 · Week 8 (final, v1.0)

CODE

```
Repository : https://github.com/jainamsethia/Curriculum-Intelligence-Engine
Tag        : v1.0
Commit     : 8807acc9d8853b26107ddd355d9af444ae76f552
Install    : pip install -e .[dev]  &&  python -m spacy download en_core_web_sm
Reproduce  : python -m curriculum_engine.run all                  (data → week2 … week8; data/processed/courses.jsonl from the
Teams zip)
Final test : python tools/final_test.py                            (clean clone → tests → run all → compare results; API
latency)
Serve / UI : uvicorn curriculum_engine.api:app --port 8000   |   streamlit run app/committee_ui.py
Slides     : cd tools && npm install && npm run slides             (architecture PNG: npm run architecture)
Folders    :
.gitignore
README.md
app
curriculum_engine/   the NLP engine (parser, data, overlap, redundancy, prerequisites, diff, Bloom, evaluation)
demo
docs/                documentation
labels/              LLM-annotated reference labels (ids + labels per pass + consensus)
legacy/              old Week 3-4 scripts and reports (history only)
legacy_mit/          old MIT OCW scripts (not used)
pyproject.toml
reports/             report templates (numbers are placeholders)
requirements.txt
results/             metrics JSON + figures per week
submissions/         per-week submission folders
tests/               unit tests (pytest)
tools/               report builder, report checker, submission packer
```

Feedback addressed

Comment	What we changed	Evidence
Week 2: wrong dataset (MIT OpenCourseWare)	NMIMS semester syllabus booklets only, from Proposal v2 on; one dataset story in every report.	`results/dataset/parse_stats.json`
Weeks 3–4: "minimal progress", "where is the code?"	Tagged code per week, one command rebuilds every result, tests, and an automated clean-clone reproduction.	`results/week8/final_testing.json`
Week 4: focus on overlapping content, missing prerequisites, redundancy	The three core capabilities are measured on a held-out split against targets fixed in advance.	`results/week8/final.json`
Weak ground truth	A documented labelling protocol with LLM-annotated reference labels, plus label-free synthetic tests and faculty Bloom tags.	`results/week8/labels.json`
Week 8 brief: final testing, report, README, presentation, demonstration, architecture diagram, final repository	Each item below, all generated from the same results files.	`submissions/week8/`, `README.md`, `docs/`

Progress vs previous week

Item	Week 7	Week 8
Testing	unit, API and UI tests; manual clean-clone gate	automated final test: 36 tests pass and a clean clone reproduces all 62 result files; latency of every endpoint
Results	one report per week	one table of the final v1.0 results against the success criteria
Architecture	figure drawn by script	Mermaid source + exported PNG, including Group 10 as a planned consumer
Presentation and demo	demo script	slides (.pptx) generated from the results files; recorded-demo checklist
README	weekly notes	final: install, run, API, data, limitations, AI use

1. Problem and what v1.0 does

Curriculum committees (Board of Studies, programme chairs) revise syllabi by reading booklets by hand. The final version (v1.0) answers their questions with evidence and a confidence score, as JSON through an API and in a committee UI: which courses overlap, which units are redundant inside a programme, which prerequisites are not taught earlier, what changed between two programmes or two years, which Bloom level each outcome targets, which emerging topics are missing, and, from job descriptions, which demanded skills are already taught and what a modernised syllabus could look like. Every output is a draft for a human decision.

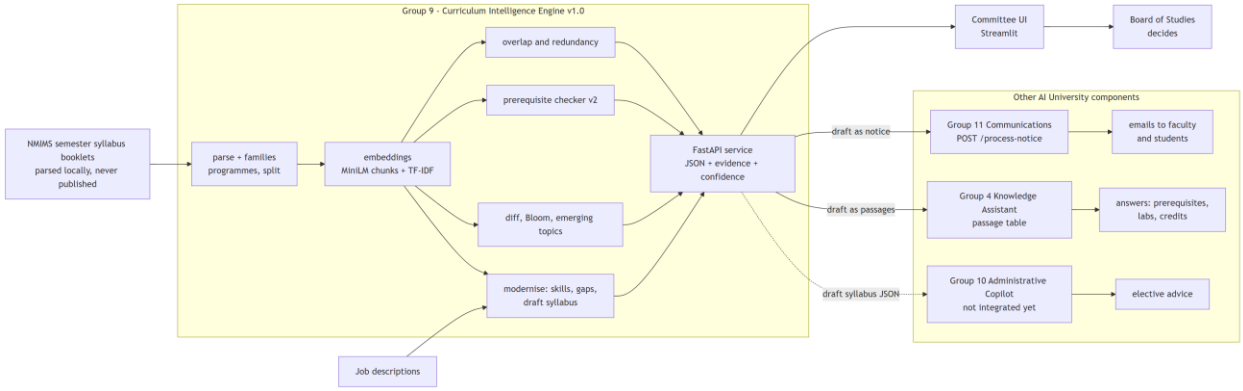
2. Data and evaluation protocol

- **Data**: NMIMS semester syllabus booklets. 531 of 2,413 non-exam PDFs have a text layer and are parsed; 1,882 are scanned. Result: 2,366 course versions in 645 course families and 13 programmes.
- **Split**: by course family (versions of one course never cross splits): 387 / 129 / 129 families, hash 4763325fd5. Thresholds are tuned on validation; the test split is used once.
- **Reference labels**: 502 items labelled by two LLM passes with different wording, without model scores, a third pass for disagreements (35 items) and 1 item excluded as uncertain:

Label set	Items	Passes 1 and 2 agree	Settled by pass 3	Uncertain (excluded)	Second labeller
course pairs (overlap 0/1/2)	200	171	28	1	Phi-3 (106), Claude (wording B) (94)
detector flags (Week 3)	187	182	5	0	Claude (wording B) (187)
prerequisite v2 flags (Week 6)	45	45	0	0	Claude (wording B) (45)
JD skill coverage (Week 7)	70	68	2	0	Claude (wording B) (70)

- **Label-free checks** run alongside: injected duplicate units, removed or renamed prerequisites, real year-to-year changes, and the 546 outcomes that carry a faculty Bloom tag.

3. System



Architecture of v1.0 (Mermaid source docs/architecture.mmd, exported with mermaid-cli). Group 10 is a planned consumer, not integrated.

Booklets are parsed locally into course records and families; MiniLM chunk embeddings and TF-IDF feed the modules (overlap by rank fusion of TF-IDF and chunk MaxSim; unit-level redundancy; prerequisite checker v2 with alias, umbrella-name, non-subject and own-content filters; curriculum diff; Bloom verb lexicon; emerging-topic coverage; curriculum modernisation from job descriptions). A FastAPI service with 9 endpoints returns JSON with evidence, confidence and the engine version; a Streamlit UI calls it; the draft syllabus is handed to Group 11 and Group 4.

4. Final results against the success criteria

Capability	Metric (test split)	Week 5	Final v1.0	Target	Met?	Final component
1 Overlap	Spearman	0.717	0.717	>= 0.6	yes	unchanged since Week 5
	ROC-AUC (any overlap)	0.965	0.965	>= 0.85	yes	unchanged since Week 5

	F1 at validation threshold	0.731	0.731	>= 0.7	yes	unchanged since Week 5
2 Redundancy	precision of top flags (reference labels)	0.800	0.800	>= 0.7	yes	unchanged since Week 5
	injected near-duplicates in top 20	1.000	1.000	>= 0.8	yes	unchanged since Week 5
3 Prerequisites	removal recall	0.880	0.711	>= 0.8	no	checker v2 (Week 6)
	precision of flags (reference labels)	0.100	0.500	>= 0.6	no	checker v2 (Week 6)
	false alarms after renaming	0.120	0.000	<= 0.2	yes	checker v2 (Week 6)
4 Curriculum diff	precision (real year pairs)	0.998	0.998	>= 0.85	yes	unchanged since Week 5
	recall (real year pairs)	0.956	0.956	>= 0.85	yes	unchanged since Week 5
5 Bloom	accuracy vs faculty tags	0.776	0.776	>= 0.7	yes	unchanged since Week 5
	within one level	0.871	0.871	>= 0.85	yes	unchanged since Week 5
6 Emerging topics	precision of flagged gaps	1.000	0.915	>= 0.7	yes	threshold 0.55 (Week 7)
7 Topic labels	share accepted	0.700	0.700	>= 0.7	yes	unchanged since Week 5
8 Curriculum modernisation	taught / gap accuracy on JD skill themes (reference labels)	–	0.857	no target set	–	new in Week 7

12 of 14 targets are met (Week 5: 13). The difference is the prerequisite checker: v2 removed the false alarms and raised flag precision, but removal recall fell below its target, and flag precision still misses its target; that precision comes from a small fresh sample, so its interval is wide. The emerging-topic row now uses the unit-match threshold chosen in Week 7 after seeing those labels, so it is optimistic. Overlap, redundancy, curriculum diff, Bloom levels and topic labels are unchanged since Week 5. The modernisation capability had no target fixed in advance.

5. Final testing

- **Clean clone** of commit `b8bf8be57e71a5beabadbdcd5b86b8e3cd006db9`: 36 passed, 7 warnings in 25.31s; `run all` finished in 381 s and reproduced every committed result (files compared: 62, reproduced: True). Machine: Windows 10.0.26200, Python 3.11.9, CPU only.
- **API on the real data** (in-process, the requests of `docs/examples`, median of 3 calls after one warm-up call that also loads the models):

Endpoint (example request)	Status	Median latency (ms)
POST /check-prerequisites (check_prerequisites)	200	10
POST /compare-curricula (compare_courses)	200	14
POST /compare-curricula (compare_programmes)	200	14
POST /detect-redundancy (detect_redundancy)	200	142
POST /emerging-topics (emerging_topics)	200	46
POST /find-overlaps (find_overlaps)	200	458
GET /health (health)	200	5
POST /map-outcomes (map_outcomes)	200	5
POST /modernize-curriculum (31 October demo)	200	255
GET /courses?q=language	200	7

- **Report and slide checks**: every number in the reports is a placeholder resolved from `results/*.json` and re-checked by `tools/check_reports.py`; the slides read the same files; the deck passed an Office XML validation.

6. Demonstration and integration

For the joint demo on 31 October our step takes 5 job descriptions on fine-tuning LLMs and 5 current NLP syllabi, finds that 3 of 10 syllabus pairs are identical, lists 15 demanded skills (6 taught, 9 gaps) and drafts a syllabus of 11 units, each naming its source. The draft goes to Group 11 as a notice in their schema (this run: mock) and to Group 4 as 17 passages in their schema; our stand-in for their offline mode answered 4 of 5 student questions from the right section. `docs/DEMO_SCRIPT.md` has the live script and a checklist for a recorded backup.

7. Responsible AI

Booklets and parsed text stay local and off GitHub; the draft syllabus is copied or templated, never generated, so nothing is invented; outputs are drafts for the Board of Studies and the UI warns against automatic course cuts; quality differs by programme (Bloom accuracy 0.692 for COMP to 0.857 for MECHATRONICS), and programmes with too little parsed data get no prerequisite checks. Details: `docs/RESPONSIBLE_AI.md`.

8. Limitations and future work

- Data: NMIMS semester syllabus booklets; 531 of 2,413 non-exam PDFs are parsed, the rest are scanned and need OCR (future work), so a course can look removed or a prerequisite missing when the evidence sits in an unread booklet.
- Labels come from LLM-annotated reference labels. No human annotators were used; results are indicative. For 94 of 200 pairs the second labeller was a Claude prompt rather than Phi-3, so agreement on those items is not independent. Detector flags (Week 3 onwards) were labelled by two Claude passes with different wording plus a pass 3, with no Phi-3, so agreement on flags is not independent. The human spot-check sheet (50 pairs) is ready but not filled in.
- Not run, and not reported as results: paraphrase variant not run (time); bge-base prerequisite re-tuning not run (time).
- The emerging-topic reference list is a draft awaiting approval; only entries whose source page was opened count as verified.
- The demo job descriptions are synthetic; Group 11's model and Group 4's knowledge base cannot be run from their repositories, so the hand-offs are checked against their schemas with a marked mock and a stand-in; Group 10 is not integrated.
- Future work: OCR for the scanned booklets, the human spot check, a faculty-confirmed prerequisite map, real job postings, and the Group 10 elective hand-off.

AI assistance

Code, report text and slides were written with AI assistance (Claude). Reference labels are LLM-annotated (Claude, Phi-3). Every number is generated from `results/*.json` by `tools/build_report.py` and verified by `tools/check_reports.py`.